RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering (Cyber Security)

PROJECT NAME: PERSONAL EXPENSE MANAGEMENT TRACKER

SUBJECT: WEB PROGRAMMING

SUBJECT CODE: CS23531

NAME: THULASI S
REG NO: 241901118

ABOUT THE DOMAIN

- Personal Finance Management, which is part of the FinTech (Financial Technology) domain
- Why it matters - It replaces manual calculations, shows where money goes, and supports better budgeting.
- About the Domain - It uses technology to help individuals record, track and manage their own money. It covers income and expense tracking, budgeting, savings awareness and financial reporting.
- It shows where money is going, so users can spend and save more carefully.

PROBLEM STATEMENT / OBJECTIVE

PROBLEM SATEMENT:

- Many people track their income and expenses in notebooks, memory or scattered notes.
- This is time-consuming and error-prone, and it gives no clear picture of spending habits, available balance or savings.
- Without organized records, it is hard to budget well or see where money is being wasted.
- There is a need for a simple, easy-to-use system that records transactions, calculates the balance automatically and presents the data clearly.

OBJECTIVE:

- Record income and expense transactions.
- Calculate the available balance automatically.
- Organize transactions by category and date.
- Display financial summaries and charts.
- Store transaction history in a MySQL database.

TOOLS, TECHNOLOGIES

TECHNOLOGY	DESCRIPTION
HTML	Builds the structure of the web pages (forms, tables, dashboard layout).
CSS	Styles the pages with colors, fonts and layout.
BootStrap	Provides ready-made responsive components, so the app looks good on any screen size.
JavaScript	Adds interactivity, such as form validation and updating the page without a reload.
PHP	Handles requests from the user, processes transactions and runs the application logic.
MYYSQL	Stores all income and expense records and retrieves them for the dashboard and history.
XAMPP	Runs Apache (web server) and MySQL on your own computer for development and testing.

IMPLEMENTATION

- Install XAMPP and start the Apache and MySQL modules.
- Keep the project files in the htdocs folder and open it at localhost/expense-tracker.
- Create a database, e.g. expense_tracker.
- Create a transactions table with these fields: id (primary key), type (income or expense), amount, category, description, date.
- HTML pages for the dashboard, add transaction form and history.
- CSS and Bootstrap for the layout, cards, tables and responsive design.
- JavaScript for form validation and interactive features.
- db.php connects to MySQL.
- history.php fetches and displays saved transactions.
- dashboard.php calculates total income, total expenses and balance (Balance = Income − Expenses) with SQL queries.
- User enters a transaction → PHP validates and processes it → MySQL stores it → the dashboard and history are refreshed with the updated data.

SOURCE CODE

functions.php

includes > functions.php

```php
<?php
declare(strict_types=1);
function h(?string $value): string {
    return htmlspecialchars($value ?? '', ENT_QUOTES, 'UTF-8');
}
function money(float $amount): string {
    return '₹' . number_format($amount, 2);
}
function getSummary(PDO $pdo, ?string $from = null, ?string $to = null): array {
    $where = [];
    $params = [];
    if ($from) { $where[] = 'transaction_date >= ?'; $params[] = $from; }
    if ($to) { $where[] = 'transaction_date <= ?'; $params[] = $to; }
    $sqlWhere = $where ? ' WHERE ' . implode(' AND ', $where) : '';
    $stmt = $pdo->prepare(
        "SELECT
            COALESCE(SUM(CASE WHEN type='income' THEN amount ELSE 0 END),0) AS income,
            COALESCE(SUM(CASE WHEN type='expense' THEN amount ELSE 0 END),0) AS expenses
        FROM transactions{$sqlWhere}"
    );
    $stmt->execute($params);
    $row = $stmt->fetch() ?: ['income' => 0, 'expenses' => 0];
    $income = (float)$row['income'];
    $expenses = (float)$row['expenses'];
    return ['income' => $income, 'expenses' => $expenses, 'balance' => $income - $expenses, 'savings' => $
}
function validDate(string $date): bool {
    $d = DateTime::createFromFormat('Y-m-d', $date);
    return $d && $d->format('Y-m-d') === $date;
}

```

SOURCE CODE

functions.php ● JS app.js ×

sets > JS app.js > ...

```
1   // Small client-side validation enhancement.
2   document.addEventListener('DOMContentLoaded', () => {
3     const form = document.getElementById('transactionForm');
4     if (!form) return;
5     form.add  const amount: number      vent => {
6       const amount = Number(document.getElementById('amount').value);
7       if (!Number.isFinite(amount) || amount <= 0) {
8         event.preventDefault();
9         alert('Please enter an amount greater than zero.');
10        document.getElementById('amount').focus();
11      }
12    });
13  });
14
```

SOURCE CODE

functions.php ● config.php ● database.sql ×

database.sql

```sql
CREATE DATABASE IF NOT EXISTS expense_tracker
  CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE expense_tracker;

CREATE TABLE IF NOT EXISTS transactions (
    id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(120) NOT NULL,
    amount DECIMAL(12,2) NOT NULL,
    type ENUM('income','expense') NOT NULL,
    category VARCHAR(60) NOT NULL,
    transaction_date DATE NOT NULL,
    notes VARCHAR(500) DEFAULT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_transaction_date (transaction_date),
    INDEX idx_type (type),
    INDEX idx_category (category)
);

-- Optional sample data. Remove these INSERT statements if you want an empty tracker.
INSERT INTO transactions (title, amount, type, category, transaction_date, notes) VALUES
('Monthly allowance', 12000.00, 'income', 'Salary / Allowance', DATE_FORMAT(CURDATE(), '%Y-%m-01'), 'Sample
('Groceries', 1800.00, 'expense', 'Food', CURDATE(), 'Sample expense'),
('Bus pass', 650.00, 'expense', 'Transport', CURDATE(), 'Sample expense'),
('Freelance work', 2500.00, 'income', 'Freelance', DATE_SUB(CURDATE(), INTERVAL 1 MONTH), 'Sample income');
```

SOURCE CODE

functions.php config.php index.php

index.php

```php
<?php
declare(strict_types=1);
require_once __DIR__ . '/config.php';
require_once __DIR__ . '/includes/functions.php';

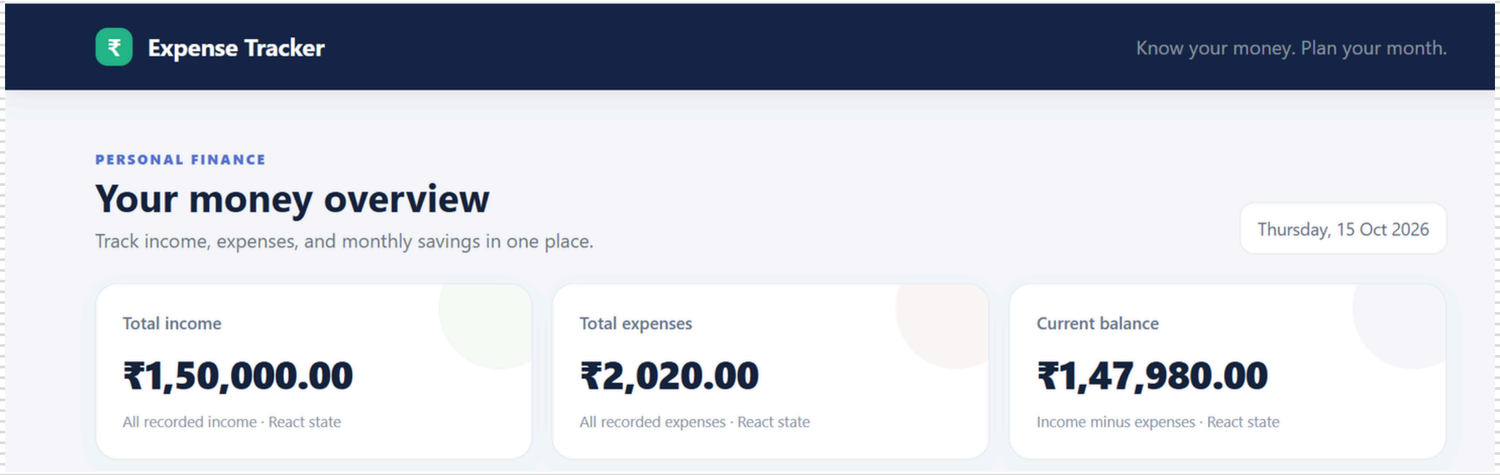
$categories = [
    'Salary / Allowance', 'Freelance', 'Business', 'Investment', 'Gift',
    'Food', 'Transport', 'Shopping', 'Bills & Utilities', 'Rent',
    'Health', 'Education', 'Entertainment', 'Travel', 'Other'
];
$errors = [];
$success = '';

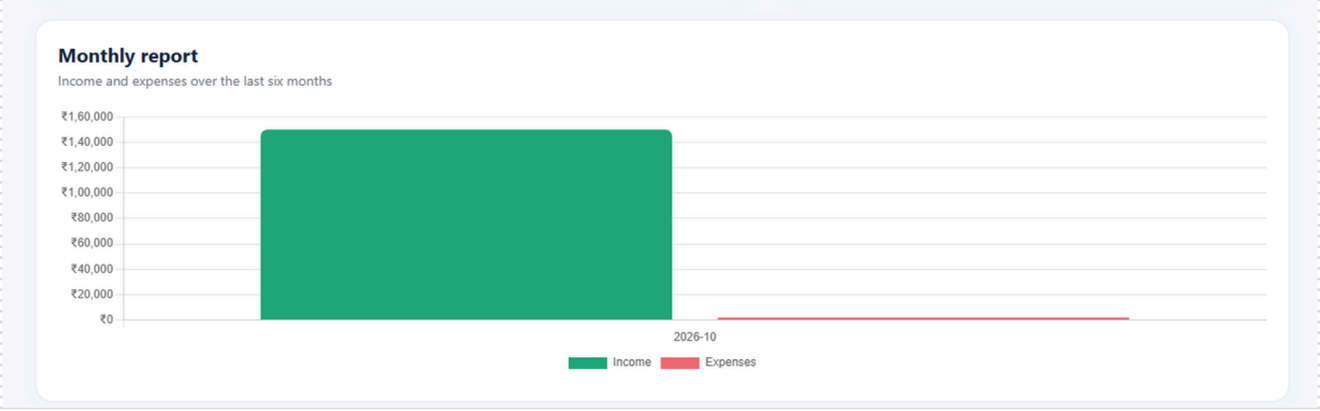
if ($_SERVER['REQUEST_METHOD'] === 'POST' && ($_POST['action'] ?? '') === 'add') {
    $title = trim((string)($_POST['title'] ?? ''));
    $amount = filter_var($_POST['amount'] ?? '', FILTER_VALIDATE_FLOAT);
    $type = (string)($_POST['type'] ?? '');
    $category = trim((string)($_POST['category'] ?? ''));
    $date = (string)($_POST['transaction_date'] ?? '');
    $notes = trim((string)($_POST['notes'] ?? ''));

    if ($title === '' || mb_strlen($title) > 120) $errors[] = 'Enter a title (maximum 120 characters).';
    if ($amount === false || $amount <= 0) $errors[] = 'Amount must be greater than zero.';
    if (!in_array($type, ['income', 'expense'], true)) $errors[] = 'Choose income or expense.';
    if (!in_array($category, $categories, true)) $errors[] = 'Choose a valid category.';
    if (!validDate($date)) $errors[] = 'Choose a valid transaction date.';
    if (mb_strlen($notes) > 500) $errors[] = 'Notes must be 500 characters or fewer.';

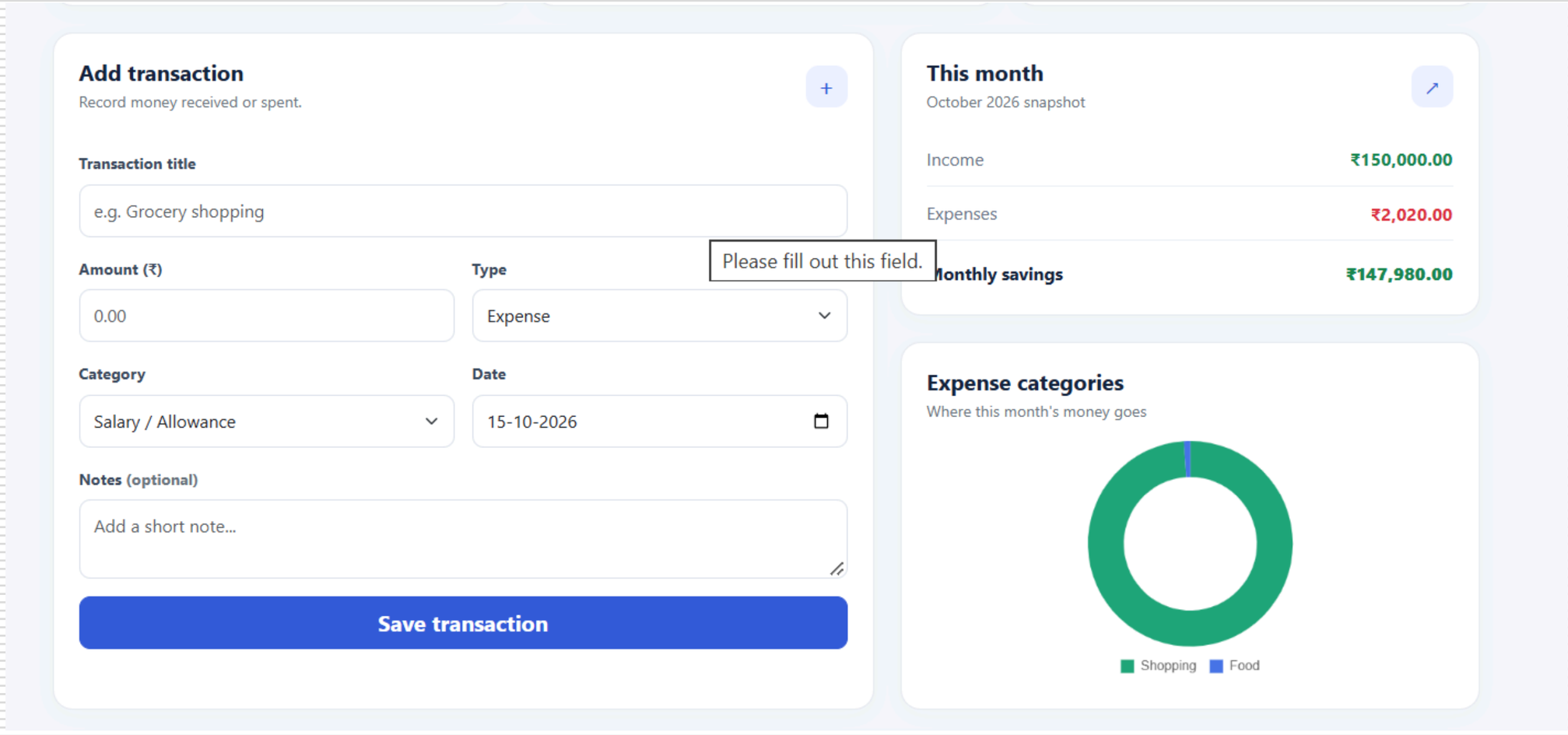
    if (!$errors) {
        $stmt = $pdo->prepare('INSERT INTO transactions (title, amount, type, category, transaction_date, not
        $stmt->execute([$title, $amount, $type, $category, $date, $notes !== '' ? $notes : null]);
        header('Location: index.php?added=1');
        exit;
    }
}
```

OUTPUT

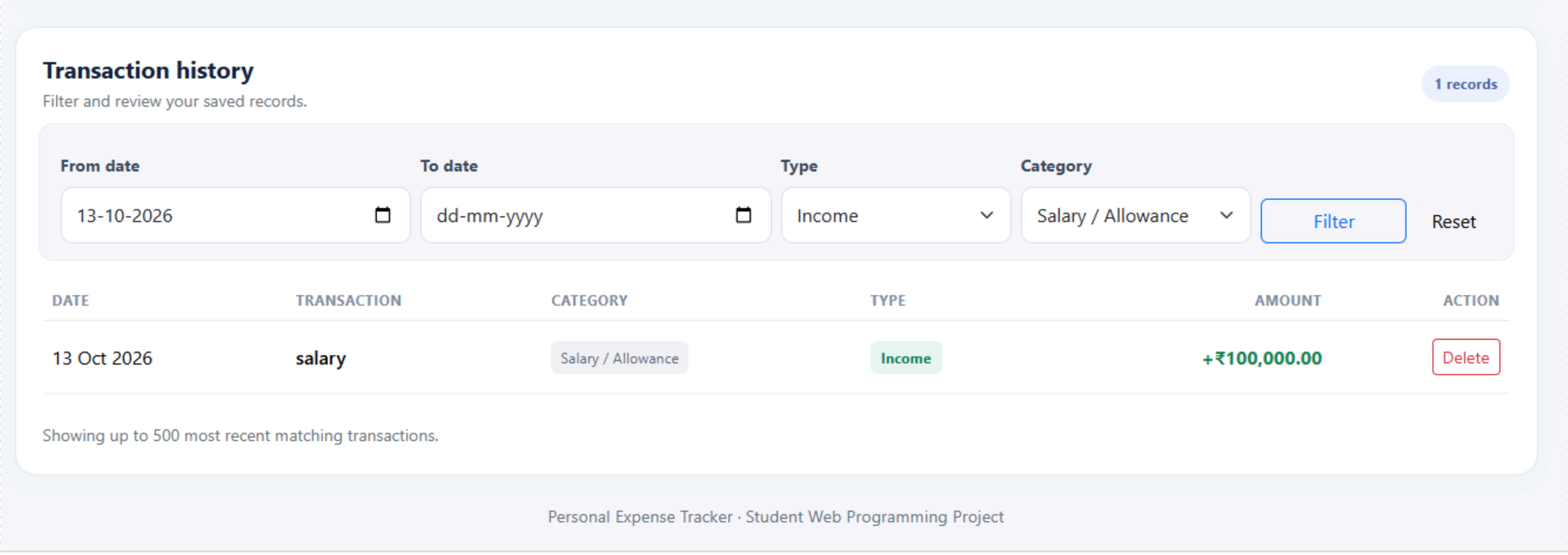

Expense Tracker
Know your money. Plan your month.
PERSONAL FINANCE
Your money overview
Track income, expenses, and monthly savings in one place.
Thursday, 15 Oct 2026
Total income
₹1,50,000.00
All recorded income · React state
Total expenses
₹2,020.00
All recorded expenses · React state
Current balance
₹1,47,980.00
Income minus expenses · React state


Monthly report
Income and expenses over the last six months
₹1,60,000
₹1,40,000
₹1,20,000
₹1,00,000
₹80,000
₹60,000
₹40,000
₹20,000
₹0
2026-10
Income
Expenses

OUTPUT


Add transaction
Record money received or spent.
+
Transaction title
e.g. Grocery shopping
Please fill out this field.
Amount (₹)
0.00
Type
Expense
Category
Salary / Allowance
Date
15-10-2026
Notes (optional)
Add a short note...
Save transaction
This month
October 2026 snapshot
↗
Income
₹150,000.00
Expenses
₹2,020.00
Monthly savings
₹147,980.00
Expense categories
Where this month's money goes
Shopping
Food

OUTPUT


Transaction history
Filter and review your saved records.
1 records
From date
13-10-2026
To date
dd-mm-yyyy
Type
Income
Category
Salary / Allowance
Filter
Reset
DATE
TRANSACTION
CATEGORY
TYPE
AMOUNT
ACTION
13 Oct 2026
salary
Salary / Allowance
Income
+₹100,000.00
Delete
Showing up to 500 most recent matching transactions.
Personal Expense Tracker · Student Web Programming Project

PROJECT OUTCOME

- Working application: A functional web app where users record income and expenses and see their financial position instantly.
- Automatic calculation: The balance is computed by the system, which removes manual errors.
- Permanent records: All transactions are stored in MySQL and can be reviewed any time.
- Better financial awareness: The dashboard and charts show spending patterns, helping users budget and save more effectively.

THANK YOU

PERSONAL EXPENSE MANAGEMENT TRACKER